

$$\begin{aligned}
 1. \text{ CM Ratio} &= \frac{\text{CM Per Unit}}{\text{Selling Price Per Unit}} \\
 &= \frac{\text{Selling Price Per Unit} - \text{Variable Cost per Unit}}{\text{Selling Price Per Unit}} \\
 &= \frac{60 - 45}{60} \\
 &= 0.25 \text{ or } 25\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Variable Expense Ratio} &= \frac{\text{Variable Cost Per Unit}}{\text{Selling Price per Unit}} \\
 &= \frac{45}{60} \\
 &= 0.75 \text{ or } 75\%
 \end{aligned}$$

$$\begin{aligned}
 2. \text{ ~~BEP~~ } \text{ BEP (Quantity)} &= \frac{\text{Total Fixed Cost}}{\text{Contribution Margin per Unit}} \\
 &= \frac{240000}{60 - 45}
 \end{aligned}$$

$$\begin{aligned}
 \text{BEP (Dollar)} &= \frac{\text{Total Fixed Cost}}{\text{CM Ratio}} = \frac{240000}{0.25} \\
 &= 960000
 \end{aligned}$$

③



Sales	100	400,000
- VC	75	300,000
CM	25	100,000
- FC	0	240,000
NOI		160,000

Increase in Sales	400,000
multiplied by CM Ratio	0.25
Expected Increase Contribution Margin	100,000

As this ~~can~~ increase in sale will not affect FC, ~~the~~ it will lead to an increase of 100,000 in NOI.

④

$$\text{BEP (Unit)} = \frac{\text{Total Fixed Cost} + \text{target profit}}{\text{cm per unit}}$$

$$= \frac{240,000 + 90,000}{15}$$

$$= 22,000$$

⑤

$$\text{margin of safety (Dollar)} = \text{Actual Sale} - \text{Break Even Sale}$$

$$= 1200,000 - 960,000$$

$$= 240,000$$

$$\text{mos \%} = \frac{\text{mos (dollar)}}{\text{Total Sales Actual}}$$

$$= \frac{240,000}{1,200,000} = 20\%$$

6(a)

$$\begin{aligned}
 DOL &= \frac{\text{Total Contribution Margin}}{\text{Total Net Operating Income}} \\
 &= \frac{\text{Total Sales} - \text{Total Variable Cost}}{\text{Total Net Operating Income}} \\
 &= \frac{1200000 - 900000}{600000} \\
 &= 5
 \end{aligned}$$

6(b)

Sales $\rightarrow 8\% \uparrow$

NOI $\rightarrow 8\% \times 5 \rightarrow 40\%$

$$\begin{aligned}
 \frac{\% \Delta \text{ in NOI}}{\% \Delta \text{ in Sales}} &= DOL \\
 &= 5 \times 8\% \\
 &= 40\%
 \end{aligned}$$

6(c)

Sales
- VC

CM

- FC

NOI

$$\begin{array}{r}
 1296000 \\
 (972000) \\
 \hline
 324000
 \end{array}$$

$$\begin{array}{r}
 324000 \\
 240000 \\
 \hline
 84000
 \end{array}$$

$$\begin{array}{r}
 84000
 \end{array}$$

$$\begin{array}{r}
 1296000 - 1200000 \\
 1200000 \\
 \hline
 \rightarrow 8\%
 \end{array}$$

$$\begin{array}{r}
 0.40 \\
 40\%
 \end{array}$$

$$\begin{array}{r}
 84000 - 60000 \\
 60000 \\
 \hline
 \rightarrow 40\%
 \end{array}$$

7(a)

Sales (24000 unit) × 60	=	Total	Per Unit	%
		1440000	60	100%
Less: VC (24000 × 48)	=	(1152000)	48	80%
			12	20%
		288000		
		(210000)		
		<u>78000</u>		

CM

- FC

Net Operating Income

$$\textcircled{b} \text{ BEP (Unit)} = \frac{\text{Total FC}}{\text{CM per Unit}}$$

$$= \frac{210000}{60 - 48} = 17500$$

$$\text{BEP (Dollars)} = \frac{\text{Total FC}}{\text{CM Ratio}} = \frac{210000}{0.20} = 1050000$$

